HACKRU FALL 2026 · REGENERON: BEST CLINICAL TRIAL SOLUTION

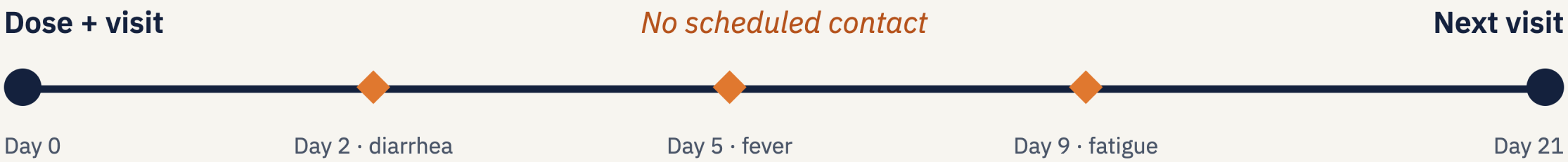
Between Visits

An AI voice agent that checks on clinical-trial patients between visits, in English or Spanish, at the patient's own pace.

bv-demo-web.onrender.com · github.com/feruzkarimovv/between-visits

THE PROBLEM

Side effects start between visits. The trial hears about them weeks later.



Symptoms surface at the next visit, from memory, or not at all. NCI built PRO-CTCAE because clinician-reported adverse events miss much of what patients actually feel.

THE PROBLEM · WHO GETS MISSED

The patients trials struggle to reach are the ones today’s tools fail.

Patients who don’t speak English

English-only staff, apps and phone lines can't hold a symptom conversation with them.

Patients who need more time

Fast voice bots and rushed calls cut off older, slower-moving patients before they finish answering.

Source: Regeneron mentor, HackRU, Oct 10, 2026

THE PROBLEM · WHY IT MATTERS

A missed symptom is a safety risk with a legal clock.

≥7

stools a day over baseline is grade 3 diarrhea. On anti-PD-1 therapy it can be the first sign of immune colitis, treated inpatient.

24 h

for the site to report a serious adverse event to the sponsor.

7 / 15 d

for the sponsor to report to FDA (fatal or life-threatening / other).

Example protocol: NCT02760498 (anti-PD-1, public on ClinicalTrials.gov)

WHAT IT DOES

From a phone call to a sponsor report in five steps.

01

Plan

Study team approves the questions (English and Spanish), the red flags, and calls on days 1, 3 and 7 after each dose.

02

Call

Voice agent says it is an AI, offers a person, and asks one approved question at a time, in the patient's language and pace.

03

Grade

Fixed CTCAE v5 rules turn answers into provisional grades. A red-flag table decides escalation. No AI grading.

04

Page

Urgent findings page the on-call nurse in seconds, before the agent's next word, and re-page until acknowledged.

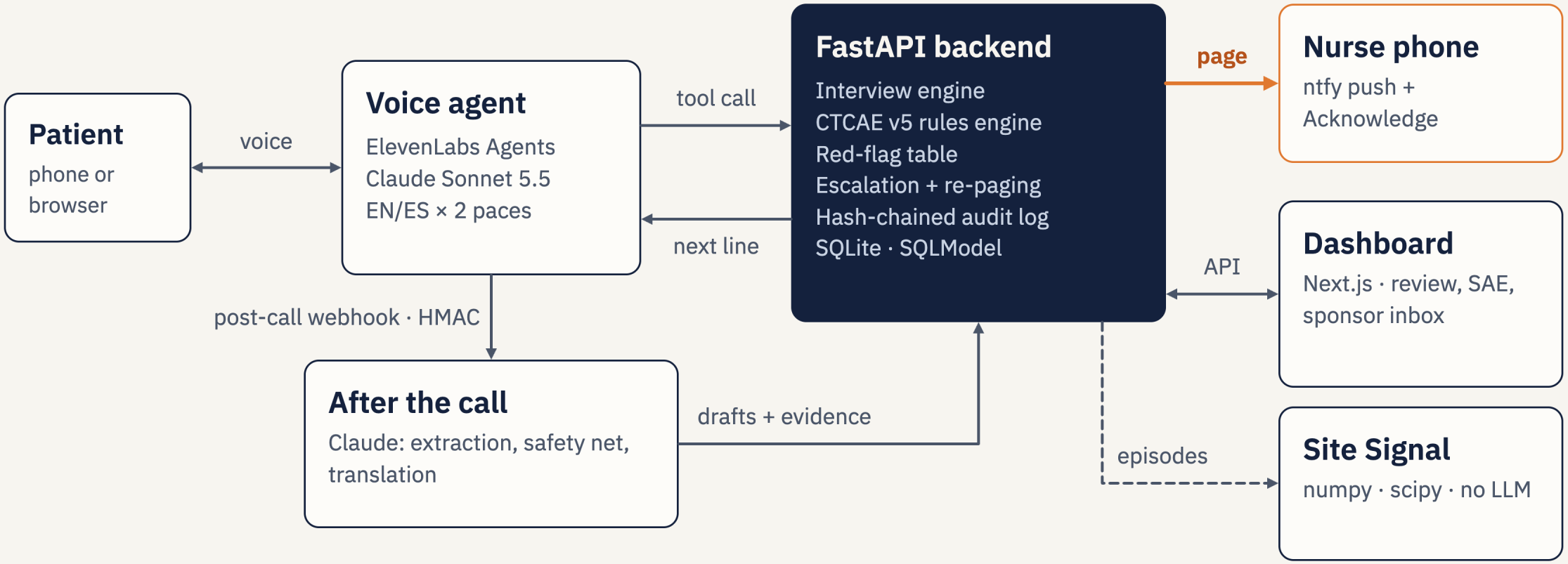
05

Report

Draft adverse-event and medication entries carry the patient's own words. Staff confirm; serious events go to the sponsor on a 24-hour clock.

TECHNICAL ARCHITECTURE

The LLM talks. The backend decides.



TECHNICAL ARCHITECTURE · ONE CALL

Rosa, Spanish, extra time: what runs where.

#	What happens	Where
1	Agent's first sentence: it is an AI, and a person is available	ElevenLabs agent
2	Rosa: "ocho veces" (8 stools a day; her baseline is 1)	Patient
3	record_answer(DIARRHEA_STOOLS_PER_DAY, 8) + her exact quote	Server tool → FastAPI
4	8 − 1 = 7 over baseline → CTCAE grade 3 + RF_DIARRHEA_SEVERE	rules.py, no LLM
5	Nurse paged before the tool returns; re-paged at 90 s, then backup	ntfy → nurse phone
6	Agent reads the approved urgent line verbatim, then moves on	data/scripts.json
7	Webhook → extraction, safety-net scan, translation → 3 drafts	Claude post-call pipeline

Safety is enforced in code, not in the prompt.

Guarantee	How the code enforces it
Rules grade, not the LLM	Grades and escalation come only from ctcae_rules.json and red_flags.json via rules.py
Approved words only	Backend builds every line from data/ in the call's language; the agent says next.say verbatim
No medical advice	Approved deferral line plus a nurse task; an LLM judge checks every eval transcript
Red flags page immediately	Page goes out before the tool returns; re-page at 90 s; backup nurse after 2 re-pages
Nothing missed after the call	Post-call safety net escalates red flags mentioned but not reported live
Only the investigator signs	Role check on seriousness, relatedness and sponsor report; anyone else gets 403
Tamper-evident record	Append-only audit log stored as a hash chain, verified at /api/audit/verify
Fail safe	Tool or rules errors escalate to a human instead of failing silently
Deployed agent = git	CI fails if the live ElevenLabs agent config hash differs from the repo

TECHNICAL ARCHITECTURE · SITE SIGNAL, FOR THE SPONSOR

Which trial sites miss what their own patients report?

Capture discordance

Episodes patients reported on calls with no match in the site's AE log, compared with other sites with the same symptom mix.

Pre-registered

Analysis plan frozen under git tag sap-v1 before any result; evaluation seed from a public drand beacon published after the tag.

Validated in simulation

20-site simulated study, 50,422 replicates. Compared with a gsm-style AE-rate KRI and a simaerep port matching R to 3.5e-15.

No LLM on this path

Flags come only from pre-specified code. A CRA approves every site query, the site answers with a code, and every step is audited.

TECH STACK

What it is built with

VOICE

ElevenLabs Agents, 4 variants
WebRTC via @elevenlabs/react
Twilio phone path

MODELS

Claude Sonnet 5.5: agent and post-call pipeline
Claude Opus 5.5: eval judge

BACKEND

Python 3.12 · FastAPI
SQLModel · SQLite
httpx · uv

FRONTEND

Next.js 16 · React 19
TypeScript · Tailwind 4
TanStack Query

CONTRACTS

OpenAPI → generated TypeScript types
Agent config as code, hash-checked

STATISTICS

numpy · scipy
Monte Carlo calibration
Clopper-Pearson intervals

ALERTS

ntfy.sh push with Acknowledge
Re-page loop + backup

OPS

Docker · Render · ngrok
GitHub Actions · Playwright
gitleaks

TESTED

Every safety rule has a test behind it.

1,052

automated tests: 661 backend, 391 statistics

29 / 29

requirements traced to passing tests

8 / 8

red flags paged across 20 simulated patients (EN/ES, both paces)

0

medical-advice responses, checked by an LLM judge

Between Visits

No symptom waits for the next visit, in any language, at any pace.

Live demo: bv-demo-web.onrender.com

Code: github.com/feruzkarimovv/between-visits